The University of British Columbia – The School of Biomedical Engineering

Deep Learning for Endometriosis Detection from Ultrasound Data

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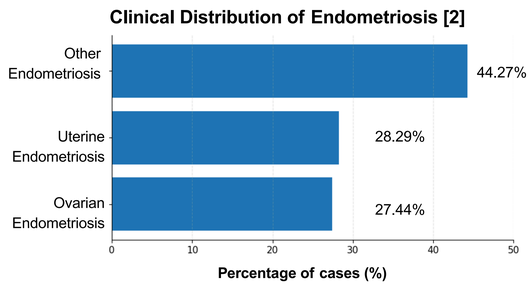
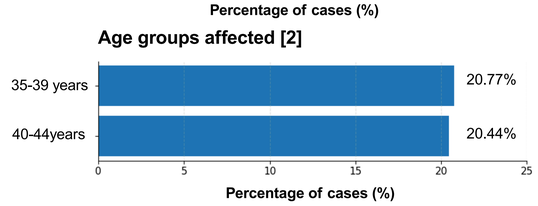
Clinical Background

Endometriosis affects 176 million women worldwide [1]





The variability in disease location and patient demographics highlights the need for robust, generalizable, and non-invasive diagnostic approaches.

Introduction

- Endometriosis diagnosis is delayed by 7–12 years on average [3]
- Gold standard (laparoscopy + biopsy) is invasive and costly. [4]
- Ultrasound was selected over MRI due to its cost-effectiveness, accessibility, and real-world clinical use. [4]
- Deep learning can improve ultrasound-based diagnosis. [1]

Objective

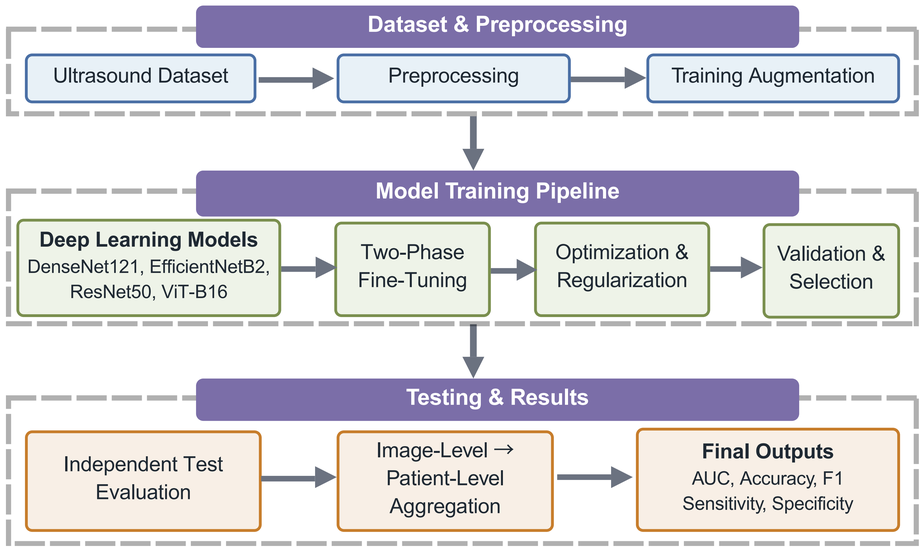
- Compare multiple deep learning architectures: DenseNet121, EfficientNet-B2, ResNet50, ViT-B16
- Evaluate both image-level and patient-level performance
- Identify the best-performing model for clinical deployment
- Develop a scalable pipeline for real-world use

Dataset Overview

- Retrospective ultrasound dataset (Rutgers, 2017–2020) [1]
- Only transvaginal ultrasound (TVUS) images used

Class	Patients	TVUS Images
Endometriosis	49	10,840
Healthy	44	7,863
Total	93	18,703

Endometriosis Detection Pipeline



Results

Comparison of Model Performance on the Test Set

Analysis Level	Model	AUC	Accuracy	F1-score	Sensitivity	Specificity
Patient Level Test	DenseNet121	**0.944**	**0.895**	**0.900**	**0.900**	**0.889**
	ViT-B16	0.856	0.684	0.700	0.700	0.667
	EfficientNet-B2	0.833	0.684	0.667	0.600	0.778
	ResNet50	0.822	0.789	0.778	0.700	**0.889**
Image Level Test	DenseNet121	**0.839**	0.615	0.516	0.354	**0.975**
	ViT-B16	0.780	**0.696**	**0.700**	**0.615**	0.807
	EfficientNet-B2	0.782	0.690	0.692	0.602	0.812
	ResNet50	0.742	0.646	0.610	0.478	0.878

Patient-Level Test Performance

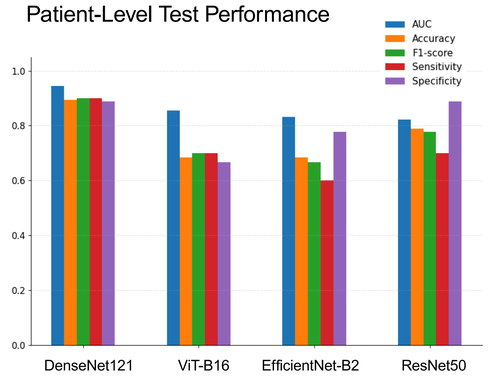
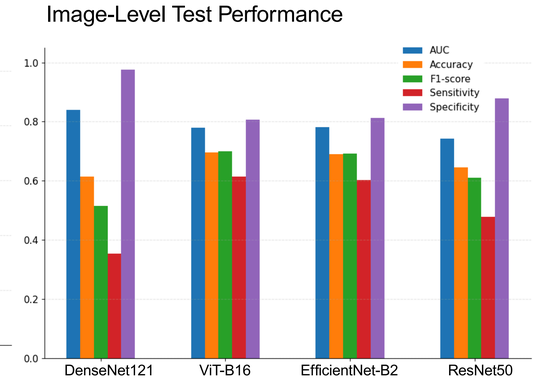
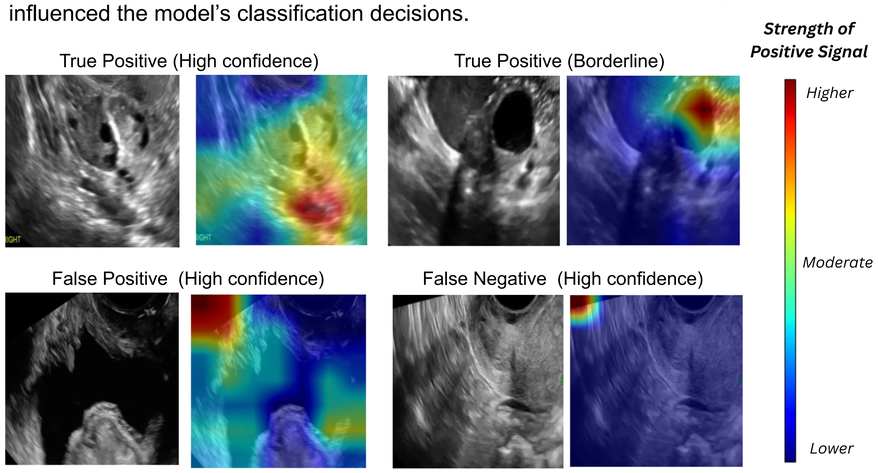


Image-Level Test Performance



Gradient-weighted Class Activation Mapping

Heatmaps were used to improve interpretability by highlighting image regions that influenced the model's classification decisions.



Conclusion

- DenseNet121 achieved the highest performance (Patient-level AUC: 0.944)
- Patient-level evaluation outperformed image-level metrics, highlighting the importance of aggregation
- Deep learning demonstrates strong potential for reliable, non-invasive endometriosis diagnosis

Future Improvements

- Validate on larger, multi-center datasets to improve generalizability
- Integrate self-supervised learning to reduce reliance on annotated data
- Develop segmentation models for lesion localization and disease staging
- Explore real-time deployment in clinical workflows

Acknowledgements:

This work was conducted on the traditional, ancestral, and unceded territory of the xʷməθkʷəy̓əm (Musqueam) People.

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Reference

[1] A. Balica et al., "Augmenting endometriosis analysis from ultrasound data using deep learning," in Medical Imaging 2023: Ultrasonic Imaging and Tomography, SPIE, Apr. 2023, pp. 118–123. Doi: 10.1117/12.2653940.
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[3] G. M. Buck Louis et al., "Incidence of endometriosis by study population and diagnostic method: the ENDO study," Fertil Steril, vol. 96, no. 2, pp. 360–365, Aug. 2011, doi: 10.1016/j.fertnstert.2011.05.087.
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